Project 2

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Q1

1 / 1

Given the points `[-1 x 1]`, what value does the condition number of the corresponding Vandermonde matrix approach as $x \to \pm 1$?

- [x] ∞
- [ ] 25
- [ ] π^2
- [ ] e^n where n is the dimension of the matrix.

Q2

1 / 1

For which values of x is `vander( [-1 x 1] )` not invertible?

- -1
- 1
- ✓ (selected) -1 and 1
- All values $x \leq -1$ and $x \geq 1$
- All values $-1 \leq x \leq 1$

Q3

1 / 1

Why is the Vandermonde matrix of a sequence of values $x_1, \ldots, x_n$ not invertible if any two x-values are equal?

- ○ Two columns of the Vandermonde matrix are identical, and thus, the matrix is not invertible.
- ✔ ● Two rows of the Vandermonde matrix are identical, and thus, the matrix is not invertible.
- ○ The columns are linearly independent, and thus the matrix is not invertible.
- ○ The rows are linearly independent, and thus the matrix is not invertible.

Q4

1 / 1

Assuming we are taking readings y_k at time t_k, why do we shift the t-values towards 0 before performing any interpolations. Note, we scale, too, but now we focus on shifting.

- [x] To reduce the condition number of the corresponding Vandermonde matrix, as the t_k values increase.
- [ ] So that the first data point occurs at 0, making the constant term equal to y_k.
- [ ] To reduce the condition number of the corresponding Vandermonde matrix, as the y_k values increase.
- [ ] To ensure that the Vandermonde matrix becomes orthogonal.
- [ ] To preserve the physical meaning of time by removing absolute timestamps.

Q5

1 / 1

Which matrix has the largest condition number?

- ○ `vander( [-1000 0 1000] )`
- ✓ ◉ `vander( [999 1000 1001] )`

Q6

1 / 1

If the condition number of a matrix A is c, what does this say about solving the system of linear of linear equations $A\mathbf{u} = \mathbf{y}$ for a given vector $\mathbf{y}$?

- [x] It indicates that relative errors in the data or arithmetic may be amplified by up to a factor of c in the computed solution $\textbf{u}$.
- [ ] It means the solution will be wrong by exactly a factor of c.
- [ ] It tells you how many iterations a numerical solver will require.
- [ ] It gives the absolute error in the solution vector $\textbf{u}$.
- [ ] If c is large, the system has no solution.

Q7

1 / 1

The condition number of the matrix `vander( [-1 0 1] )` is approximately \$3.2255\$. The condition number is minimized with the matrix `vander( [-1.106681919 0 1.106681919] ) )` where the condition number is approximately \$3.1463\$. Does it make any sense to map and scale to these new values?

- [x] No, because the improvement in conditioning is marginal (about a few percent), so it is unlikely to produce a measurable improvement in the final result compared with other error sources (sensor noise, quantization, model mismatch, and the rest of the floating-point pipeline).
- [ ] Yes, because any reduction in condition number is automatically worth implementing.
- [ ] No, because scaling and shifting change the physical meaning of time and therefore invalidate the data.
- [ ] No, because a condition number near 3 means the matrix is singular or nearly singular.

Q8

1 / 1

This code finds the interpolating quadratic that passes through the three points $(1, 4.8)$, $(2, 4.2)$, and $(3, 4.1)$, then differentiates the polynomial, finds the root of that polynomial, and evaluates the original polynomial at that point:

```
p = polyfit( [1 2 3], [4.8 4.2 4.1], 2 )
dp = polyder( p )
x = roots( dp )
polyval( p, x )
```

Executing this, you find that the minimum is close to $(2.7, 4.0775)$.

Next, do the same with the following:

```
x1 = 1.100; x2 = 1.105; x3 = 1.110;
c1 = 3.146547025441603
c2 = 3.146282197679608
c3 = 3.146333441416843
p = polyfit( [x1 x2 x3], [c1 c2 c3], 2 )
dp = polyder( p )
x = roots( dp )
polyval( p, x )
```

What is the relative error of x2 to the value 1.106681919, and what is the **relative** error of the minimum of the interpolating quadratic is as an approximation to 1.106681919?

- [x] 0.001519785 and 0.000006728
- [ ] 0.1519785 and 0.0006728
- [ ] 0.001681919 and 0.00000745
- [ ] 0.1681919% and 0.000745%

Q9

1 / 1

Why in the previous question is the the approximation found using the interpolating quadratic so much better?

- [x] Because the quadratic interpolation effectively uses curvature information from all three nearby samples, canceling first-order error and yielding a second-order accurate estimate of the minimizer.
- [ ] Because MATLAB's `polyfit` internally uses higher precision arithmetic than ordinary floating-point.
- [ ] Because the interpolating quadratic exactly reproduces the true condition-number curve.
- [ ] Because taking the derivative removes numerical noise, automatically improving accuracy.

Q10

1 / 1

Which has the largest condition number?

- ( ) `vander( [-10 0 10] )`
- ( ) `vander( [-1 0 1] )`
- (•) `vander( [-0.1 0 0.1] )` ✓

Q11

1 / 1

What does the polynomial appear to converge to as h becomes small?

```
for n = 1:5
  h = 10^-n;
  polyfit( [-h, 0, h], [sin(1 - h), sin(1), sin(1 + h)],
end
```

- [x] It converges to the second-order Taylor expansion of $sin(1 + h)$.
- [ ] It converges to $sin(1 + h)$.
- [ ] It converges to the exact sine function because three points uniquely determine it.
- [ ] It converges to zero because the interval $[-h, h]$ shrinks to a point.
- [ ] It converges to a constant equal to $sin(1)$.

Q12

1 / 1

Numerically, would the approximations in the previous question converge if we used, for example, `for n = 1:16`?

- ✔ (selected) Not reliably: for large n, h becomes so small that $\sin(1\pm h)$ is numerically indistinguishable from $\sin(1)$ and the Vandermonde system becomes ill-conditioned, so roundoff dominates and the coefficients stop improving.
- Yes, because the Taylor limit exists analytically, so floating-point computations must also converge for all n.
- No, because polynomial interpolation of $\sin$ is mathematically impossible.
- Yes, because `polyfit` uses symbolic arithmetic and therefore avoids roundoff error.
- Yes, because as $h\to 0$ the condition number of the Vandermonde matrix approaches one, guaranteeing numerical stability.

Q13

1 / 1

What is the real root of the polynomial $x^5 - 5.1x^4 + 16.45x^3 - 2.045x^2 + 1.4724x - 0.14274$?

- [x] The real root is $x = 0.1$.
- [ ] The real root is $x = -0.14274$.
- [ ] The real root is $x = 10$.
- [ ] The real root is $x = 1$.
- [ ] The real root is $x = 0.10052634543588832$.